LCC Pico Board Overview

Documentation, source and gerbers for boards may be found at: https://github.com/bocabob/LCC-RPi-Pico-Board

Evolution

The LCC Pico development boards evolved from LocoNet designs dating back to 2021. These early turntable boards used Arduino Nanos and depended on turnout commands for indexing to hard-coded stepper positions. Fascia panel input was desired so a keypad version was developed. This proved to be cumbersome, so a small touch screen was added with a Mega processor. The screen size needed to increase to nine inches for usability so the Raspberry Pi Pico processor was selected. For me, the Pico represents the 'Goldilocks' processor of very good function, technical stability, and the easy-to-use Arduino IDE. Moving to the Pico enabled LCC migration in 2024 when the Single Thread library began to support the RP2040 processor family. With the OpenLcbCLib, full protocol support was enabled.

Families

There are two basic designs, the Node and the Stepper. The Stepper provides for a stepper motor driver and dedicates supporting GP pins. The Node provides 16 GP pins through two 10P IDC connectors. While it will work with either the Pico or Pico2, the latter is recommended due to enhanced capabilities.

Common Features

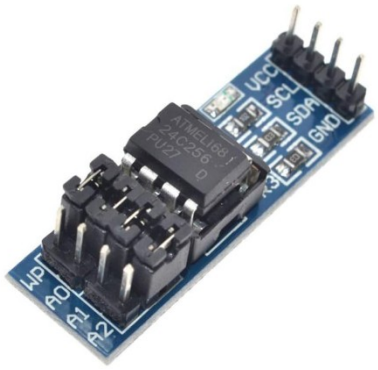
Power

The boards provide dual switching power output of 3.3V and 5V with input either from the LCC CAN bus or an external power source between 8 and 15 volts. The boards may be configured to inject power to the CAN bus or to isolate power for either or both of the CAN bus jacks. Auxiliary power outputs for both 3.3V and 5V are available on the board.

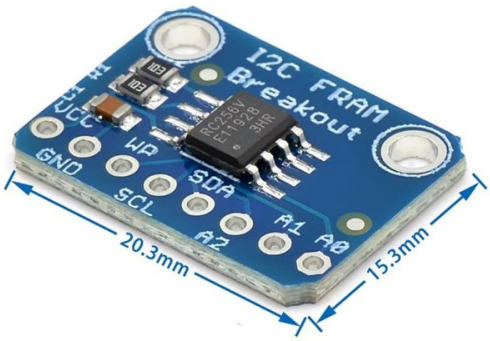
CAN Bus Control

While there is an early design that uses the MCP251863 integrated CAN controller and transceiver, later designs settled on the MCP2518 and MCP2562 discrete hardware components (chips) for the LCC interface. Both work fine, it is just easier to hand solder the discrete chips.

Non-Volatile Memory (NVM)



I2C nonvolatile memory should be used to reduce strain on the Pico flash. External NVM may be attached to either a I2C header, the FRAM Breakout header, or soldered down to the IC5 footprint. If either of the latter two choices are used, the I2C Address jumpers may be used to select non-default addresses or implement write protection. Otherwise, these would be functions of the module plugged into the I2C header.



A typical part would be a module utilizing an AT25C256 chip (left) or RC256V FRAM (right).

Parts compatible with the following should work for soldering to the footprint on the board.

- BR24G512F-3AGTE2 I2C BUS(2-Wre) 512K or other sizes with equivalent pinouts
- MB85RC256VPF-G-BCERE1 256kbit FRAM with I2C

I2C

I2C is used to address NVM. One or both of the available I2C channels on the Pico have headers on the board for external devices. Care should be taken should you attach peripherals to the same channel as the NVM lest there be interference. Some I2C devices will require 5 volts and so there is selection via jumper between the two available voltages.

Buttons

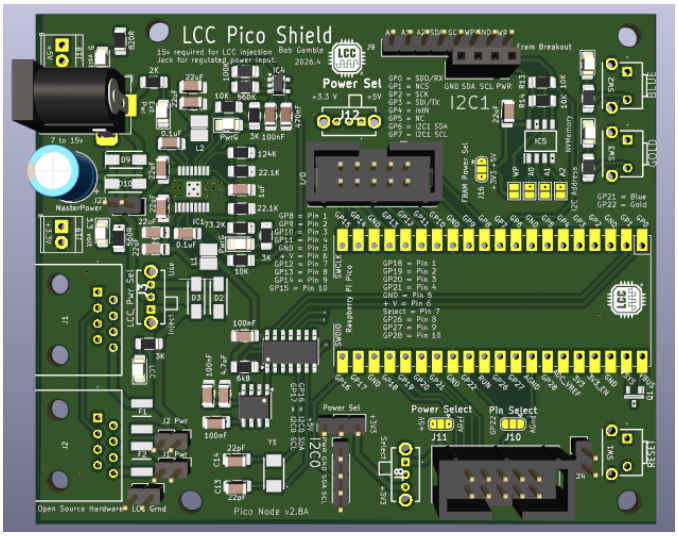
As there is no reset button on the Pico, one is provided on the board. There is also a header so that a button may be wired in for remote reset. There is also provision for blue and gold buttons should these be desired. Later designs have LEDs for these two buttons.

IDC Connections

GP pins and other connections are available via 10P IDC connectors. Pinouts may vary but are printed on the board and in the respective board user guide. Power is provided on pins 5 & 6 and is typically selectable except for the Turntable connector on the Stepper board.

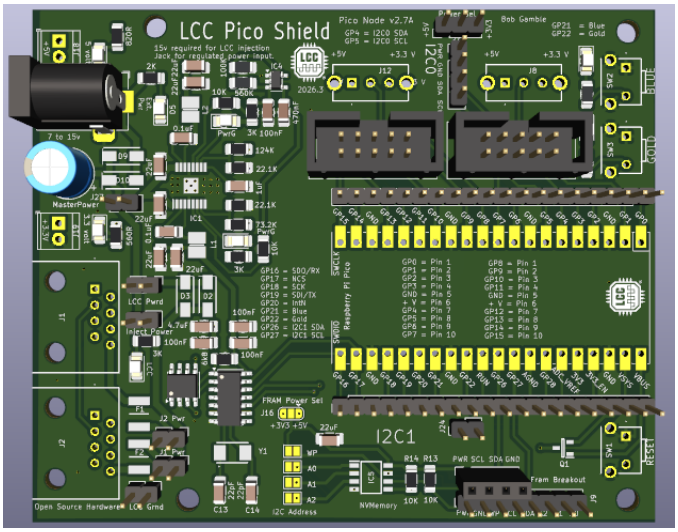
The higher GPIO pins are used on the 2.8 Node board for the second IDC connector making the analog pins usable for node peripherals.

Node 2.8



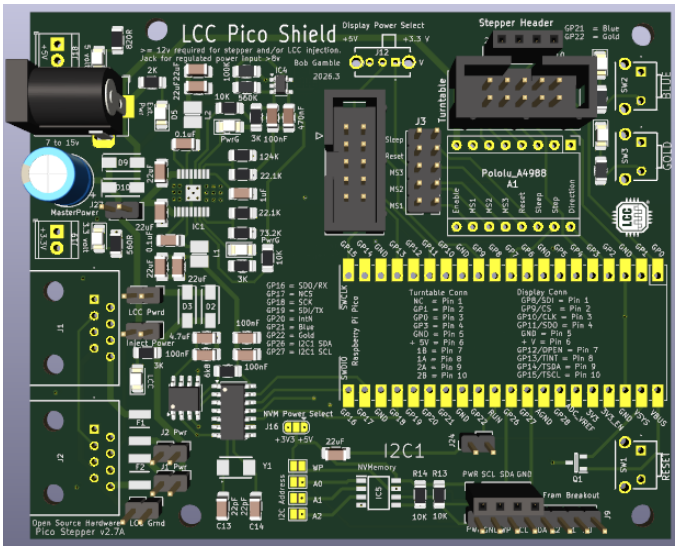
This node exposes the analog GPIO pins as an alternate implementation. Other than that, it has essentially the same functionality as Node 2.7.

Node 2.7



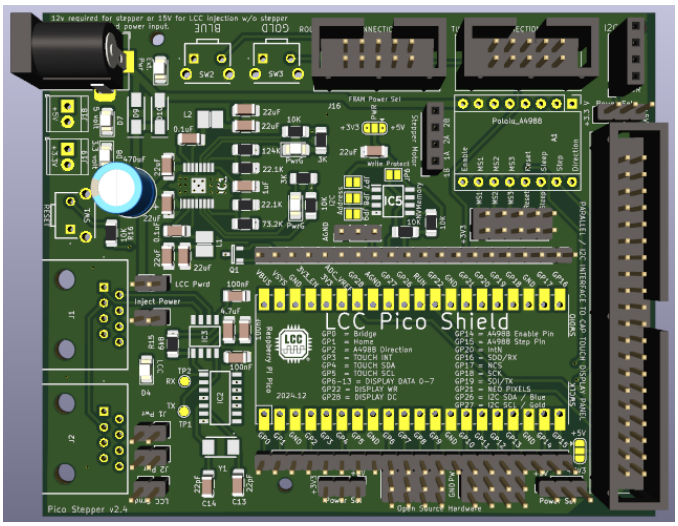
This node drives two 10P IDC connectors with a total of 16 GPIO pins. It is intended to drive servos and lights or to take inputs from detectors or buttons.

Stepper 2.7



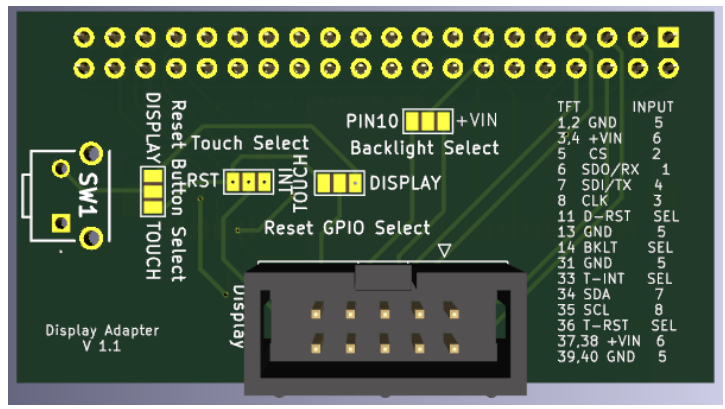
This node has provision for a stepper motor driver and is intended to control a turntable and SPI connected display.

Stepper 2.4

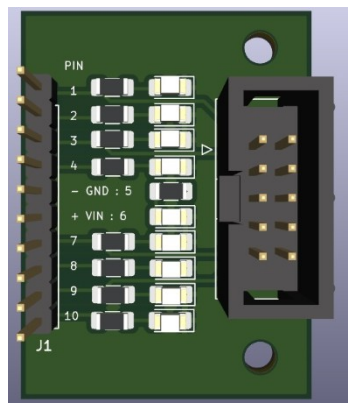


This node has provision for a stepper motor driver and is intended to control a turntable and parallel connected display.

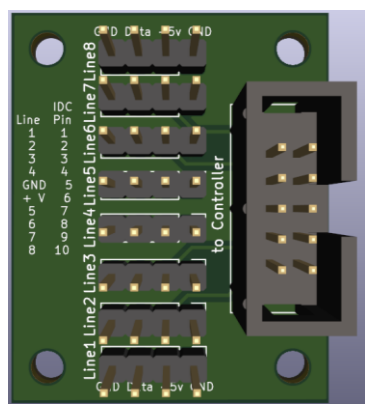
IDC Breakout Boards



Display adapter attaches to a BuyDisplay ER-TFTM101 10.1” touchscreen for SDI display control and I2C touch. There is provision for alternate assignments of line 10 and a reset button.



This IDC breakout has optional LEDs for testing.



This IDC breakout has power and ground pins around the GPIO pin to facilitate connections to inputs, servos, NeoPixels, or another device.

Board Pin Assignments

				2.5 Node board	2.6 Node board	2.7 Node board	2.8 Node board	2.4 Stepper board	2.7 Stepper board
1	gp0	SPI0 RX	I2C0 SDA	header pin	header pin	I/0-1:Pin1	ACAN_RX_PIN	Bridge sensor	TT-Pin3 - Bridge sensor
2	gp1	SPI0 CSN	I2C0 SCL	header pin	header pin	I/0-1:Pin2	ACAN_CS_PIN	Home sensor	TT-Pin2 - Home sensor
3	GND								
4	gp2	SPI0 SCK	I2C1 SDA	Npixel 1	Npixel 1	I/0-1:Pin3	ACAN_CSK_PIN	Stepper Direction	Stepper Direction
5	gp3	SPI0 TX	I2C1 SCL	Npixel 2	Npixel 2	I/0-1:Pin4	ACAN_TX_PIN	Touch Interrupt	TT-Pin4 - NeoPixel data line
6	gp4	SPI0 RX	I2C0 SDA	I2C0 SDA	I2C0 SDA	I/0-1:Pin7 / I2C0 SDA	ACAN_INT_PIN	Touch SDA	header pin
7	gp5	SPI0 CSn	I2C0 SCL	I2C0 SCL	I2C0 SCL	I/0-1:Pin8 / I2C0 SCL	header pin	Touch SCL	header pin
8	GND								
9	gp6	SPI0 SCK	I2C1 SDA	Npixel 3	Npixel 3	I/0-1:Pin9	I2C1 SDA	Display Data 0	Stepper Enable
10	gp7	SPI0 TX	I2C1 SCL	Npixel 4	Npixel 4	I/0-1:Pin10	I2C1 SCL	Display Data 1	Stepper Step
11	gp8	SPI1 RX	I2C0 SDA	I/0-1:Pin10	I/O - Pin10	I/0-2:Pin1	I/0-1:Pin1	Display Data 2	I/0-2:Pin1 : SPI1 SDI
12	gp9	SPI1 CSn	I2C0 SCL	I/0-1:Pin9	I/O - Pin9	I/0-2:Pin2	I/0-1:Pin2	Display Data 3	I/0-2:Pin2 : SPI1 CS
13	GND								
14	gp10	SPI1 SCK	I2C1 SDA	I/0-1:Pin8	I/O - Pin8	I/0-2:Pin3	I/0-1:Pin3	Display Data 4	I/0-2:Pin3 : SPI1 CLK
15	gp11	SPI1 TX	I2C1 SCL	I/0-1:Pin7	I/O - Pin7	I/0-2:Pin4	I/0-1:Pin4	Display Data 5	I/0-2:Pin4 : SPI1 SDO
16	gp12	SPI1 RX	I2C0 SDA	I/0-1:Pin4	I/O - Pin4	I/0-2:Pin7	I/0-1:Pin7	Display Data 6	I/0-2:Pin7 : Touch I2C0 SDA
17	gp13	SPI1 CSn	I2C0 SCL	I/0-1:Pin3	I/O - Pin3	I/0-2:Pin8	I/0-1:Pin8	Display Data 7	I/0-2:Pin8 : Touch I2C0 SCL
18	GND								
19	gp14	SPI1 SCK	I2C1 SDA	I/0-1:Pin2	I/O - Pin2	I/0-2:Pin9	I/0-1:Pin9	Stepper Enable	I/0-2:Pin9 : Touch INT
20	gp15	SPI1 TX	I2C1 SCL	I/0-1:Pin1	I/O - Pin1	I/0-2:Pin10	I/0-1:Pin10	Stepper Step	I/0-2:Pin10 : open
21	gp16	SPI0 RX	I2C0 SDA	ACAN_RX_PIN	ACAN_RX_PIN	ACAN_RX_PIN	I2C0 SDA	ACAN_RX_PIN	ACAN_RX_PIN
22	gp17	SPI0 CSn	I2C0 SCL	ACAN_CS_PIN	ACAN_CS_PIN	ACAN_CS_PIN	I2C0 SCL	ACAN_CS_PIN	ACAN_CS_PIN
23	GND								
24	gp18	SPI0 SCK	I2C1 SDA	ACAN_CSK_PIN	ACAN_CSK_PIN	ACAN_CSK_PIN	I/0-2:Pin1	ACAN_CSK_PIN	ACAN_CSK_PIN
25	gp19	SPI0 TX	I2C1 SCL	ACAN_TX_PIN	ACAN_TX_PIN	ACAN_TX_PIN	I/0-2:Pin2	ACAN_TX_PIN	ACAN_TX_PIN
26	gp20		I2C0 SDA	ACAN_INT_PIN	ACAN_INT_PIN	ACAN_INT_PIN	I/0-2:Pin3	ACAN_INT_PIN	ACAN_INT_PIN
27	gp21		I2C0 SCL	Blue Button	Blue Button	Blue Button	I/0-2:Pin4 & Blue Button	NeoPixel data line	Blue Button
28	GND								
29	gp22			Gold Button	Gold Button	Gold Button	I/0-2:Pin7 & Gold Button	Display WR	Gold Button
30	RUN			Reset Button / header	Reset Button / header	Reset Button / header	Reset Button / header	Reset Button	Reset Button / header
31	gp26	ADC0	I2C1 SDA	I2C1 SDA	I2C1 SDA	I2C1 SDA	I/0-2:Pin8	I2C SDA / Blue Button	I2C1 SDA
32	gp27	ADC1	I2C1 SCL	I2C1 SCL	I2C1 SCL	I2C1 SCL	I/0-2:Pin9	I2C SCL / Gold Button	I2C1 SCL
33	GND	A GND		header pin	header pin	header pin	I/0-2:Pin7 alternative		header pin
34	gp28	ADC2		header pin	header pin	header pin	I/0-2:Pin10	Display D/C	header pin
35		ADC_VREF		header pin	header pin	header pin	I/0-2:Pin6 alternative		header pin
36	3v3_OUT								
37	3V3_EN								
38	GND								
39	VSYS								
40	VBUS								
									TT-Pin1 - open
	GND			I/0-1:Pin5	I/0-1:Pin5	I/0-1:Pin5	I/0-1:Pin5		I/0-1:Pin5
	Vselect1			I/0-1:Pin6	I/0-1:Pin6	I/0-1:Pin6	I/0-1:Pin6		I/0-1:Pin6
	GND					I/0-2:Pin5			I/0-2:Pin5
	Vselect2					I/0-2:Pin6			I/0-2:Pin6

Software

Example node applications may be found in:

https://github.com/bocabob/LCC_RPiPico_Turntable

https://github.com/bocabob/LCC_RPiPico_Roundhouse

https://github.com/bocabob/LCC_RPiPico_PixelLights

Board:

Must use the RP2040 board library by Earle Philhower: https://arduino-pico.readthedocs.io/en/latest/install.html

LCC support:

https://github.com/openlcb/OpenLCB_Single_Thread

https://github.com/JimKueneman/OpenLcbCLib

CAN Hardware:

ACAN2517 Arduino library.

External Storage:

I2C_EEPROM Arduino library by Rob Tillaart

FRAM_I2C Arduino library by Rob Tillaart

NeoPixel:

NeoPixelBus by Makuna Arduino library (preferred)

NeoPixelConnect Arduino library by Alan Yorinks.

Other preferred libraries for the RPi Pico:

Display

#include <TFT_eSPI_RA8876.h> // display library [modified] https://github.com/bocabob/TFT_eSPI_RA8876

#include <RA8876_RP2040.h> // display library [modified] https://github.com/bocabob/RA8876_RP2040

#include "my_bb_captouch.h" //capacitive touch driver [modified] https://github.com/bocabob/my_bb_captouch

Servos

#include <ServoEasing.hpp> // great library for getting slow servo action, including bounces

#include <PCA9685_servo_driver.h>

#include <PCA9685_servo.h>

Stepper Motor

#include "AccelStepper.h" // by Mike McCauley [modified]